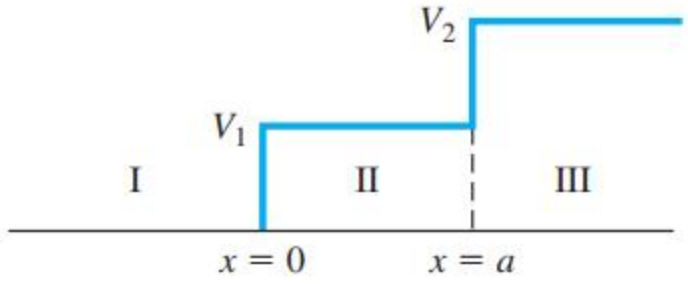
Homework 3 Due on 3/25/2026

Note:

- You may need to use Matlab or other software for this homework.
- You are not allowed to copy from others' solution or any solution on line.

1. A particle is described in 1-D by a wavefunction: $\Psi = Be^{-2x}$ for $x \geq 0$ and $\Psi = Ce^{4x}$ for $x \leq 0$, and B and C are real constants.
   a) Calculate B and C to make Ψ a valid wavefunction. (5 point)
   b) Where is the particle most likely to be? (5 point)
   c) What is the expectation of the position and momentum this states? (5 point)

2. A potential function is shown in Figure below with V_1=0.5eV, V_2 = 1eV and a=10nm, with incident particles coming from $-\infty$ with a total energy of E. Please calculate and plot the transmission coefficient versus energy E (15 point)



3. Consider a free electron bound within a two-dimensional infinite potential well defined by V=0 for $0 < x < 4\text{nm}$, $0 < y < 3\text{nm}$, and $V = \infty$ elsewhere.
   a) Determine the expression for the allowed electron energies. (10 point)
   b) Describe any similarities and any differences with the results of the one-dimensional infinite potential well. (5 point)

4. Delta function $\delta(x)$ has the following properties.

$\delta(x) = \begin{cases} \infty, & x = 0 \\ 0, & x \neq 0 \end{cases}$, and $\int_{-\infty}^{+\infty} \delta(x)dx = 1.$

Please prove $\int_{-\infty}^{+\infty} \delta(x - x_0)f(x)dx = f(x_0)$ for any f(x). (5 point)

$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx} \\ te^{ikx} \end{cases}$ $\psi(0) = \psi(0) \Rightarrow 1+r=t$

$\frac{\hbar^2}{2m}(\psi'(0) - \psi'(0)) + V_0\psi(0) = 0$

$\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\cdot\psi(x)$

$\frac{ik\hbar^2}{2m}(t-1+r) + 1+r = 0$

5. Consider a system with the potential $V(x) = V_0\,\delta(x)$, where $V_0>0$ and $\delta(x)$ is the delta function.

$|J_{in}| = \left|Re\left\{\psi\cdot\frac{\hat{p}}{m}\psi^*\right\}\right| = \left|Re\left\{e^{ikx}\cdot\left(\frac{-i\hbar}{m}\right)\cdot(-ik)\cdot e^{-ikx}\right\}\right| = \frac{\hbar}{m}Re\{k\}$ $|J_T| = |t|^2\cdot\frac{\hbar}{m}Re\{k\}$. $T = |t|^2$

a) Write the wave solutions that apply in each region and the corresponding boundary conditions. (10 point)

b) Please derive the transmission coefficient T as a function of electron energy E. (10 point)

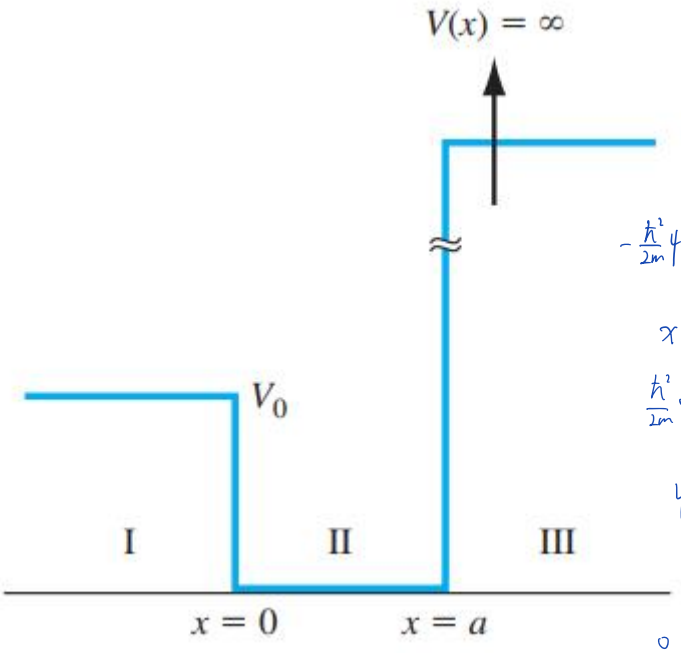
c) Please plot T vs E with Matlab. Assume electron mass m = 0.01×9.1e-31kg, V_0= 0.1eV. (10 point)

6. Consider the one-dimensional potential function shown in Figure below. Assume the total energy of an electron is $E < V_0$.

a) Write the wave solutions that apply in each region. (5 point)

b) Write the set of equations that result from applying the boundary conditions. (5 point)

c) If V_0=0.5eV, and a= 5nm, please find all the eigen energy of the bound state of the system. (10 point)



$-\frac{\hbar^2}{2m}\psi'' + V_0\psi = E\cdot\psi$

$x < 0$

$\frac{\hbar^2}{2m}\psi'' - (V_0-E)\psi = 0$

$\psi = Ae^{kx} + Be^{-kx}$ $k = \frac{\sqrt{2m(V_0-E)}}{\hbar}$

束縛態 $\longrightarrow \psi(x) = Ae^{kx}$

$0 \sim a$ ...

$> a$ ...

1. $\psi(x) = \begin{cases} B\cdot e^{-2x}, & x \geq 0 \\ Ce^{4x}, & x \leq 0 \end{cases}$

(a) $\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1$

$\Rightarrow \int_{-\infty}^{0} |\psi(x)|^2 dx + \int_{0}^{+\infty} |\psi(x)|^2 dx = 1$

$\int_{-\infty}^{0} C^2 e^{8x} dx + \int_{0}^{+\infty} B^2 e^{-4x} dx = 1$

$\Rightarrow \frac{C^2}{8} + \frac{B^2}{4} = 1$

$\psi(0^+) = \psi(0^-) \Rightarrow B = C$

Since $\psi(x) \geq 0$, B and C are real.

we have $B = C = \frac{2\sqrt{6}}{5} \approx 1.633$

(b) when $x \geq 0$

$\psi(x) = \frac{2\sqrt{6}}{5} \cdot e^{-2x}$

$\frac{d|\psi(x)|^2}{dx} = -\frac{16}{5} e^{-4x} < 0$, $|\psi(x)|^2$ maximized at $x = 0$

when $x \leq 0$

$\psi(x) = \frac{2\sqrt{6}}{5} e^{4x}$

$\frac{d|\psi(x)|^2}{dx} = \frac{32}{5} e^{8x} > 0$, $|\psi(x)|^2$ maximized at $x = 0$

Thus the particle most likely to be at $x = 0$.

(c) $\psi(x) = \begin{cases} \frac{2\sqrt{6}}{5} e^{-2x}, & x \geq 0 \\ \frac{2\sqrt{6}}{5} e^{4x}, & x \leq 0 \end{cases}$

$\langle p \rangle = \int \psi^*(x) \cdot (-i\hbar \frac{\partial}{\partial x}) \cdot \psi(x) dx$

$= \int_{0}^{+\infty} \frac{2\sqrt{6}}{5} e^{-2x} \cdot (-i\hbar \cdot \frac{\partial}{\partial x}) \frac{2\sqrt{6}}{5} e^{-2x} dx + \int_{-\infty}^{0} \frac{2\sqrt{6}}{5} e^{4x} \cdot (-i\hbar \cdot \frac{\partial}{\partial x}) \frac{2\sqrt{6}}{5} e^{4x} dx$

$= \frac{4i\hbar}{5} - \frac{4i\hbar}{5}$

$\Rightarrow \langle p \rangle = 0$

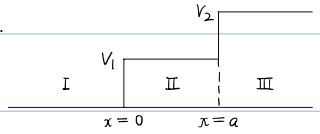
$\langle x \rangle = \int \psi_{(x)}^* \cdot \hat{x} \cdot \psi(x) dx$

$= \int_{0}^{+\infty} \frac{2\sqrt{6}}{5} e^{-2x} \cdot x \cdot \frac{2\sqrt{6}}{5} e^{-2x} dx + \int_{-\infty}^{0} \frac{2\sqrt{6}}{5} \cdot e^{4x} \cdot x \cdot \frac{2\sqrt{6}}{5} \cdot e^{4x} dx$

$\langle x \rangle = \frac{1}{6} - \frac{1}{24}$

$\Rightarrow \langle x \rangle = \frac{1}{8}$

2.



Assume wave function $\psi(x) = \begin{cases} e^{ik_1x} + r_1 e^{-ik_1x} & x<0 \\ t_1 e^{ik_2x} + r_2 e^{-ik_2x} & 0<x<a \\ t_2 e^{ik_3x} & a<x \end{cases}$

$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x)$

$\Rightarrow \begin{cases} x<0: & E = \frac{\hbar^2 k_1^2}{2m}, \quad k_1 = \frac{\sqrt{2mE}}{\hbar} \\ 0<x<a: & E = \frac{\hbar^2 k_2^2}{2m} + V_1, \quad k_2 = \frac{\sqrt{2m(E-V_1)}}{\hbar} \\ x>a: & E = \frac{\hbar^2 k_3^2}{2m} + V_2, \quad k_3 = \frac{\sqrt{2m(E-V_2)}}{\hbar} \end{cases}$

$\begin{cases} \psi(0^-) = \psi(0^+) \\ \psi'(0^-) = \psi'(0^+) \\ \psi(a^-) = \psi(a^+) \\ \psi'(a^-) = \psi'(a^+) \end{cases} \Rightarrow \begin{cases} 1 + r_1 = t_1 + r_2 \\ k_1(1 - r_1) = k_2(t_1 - r_2) \\ t_1 e^{ik_2a} + r_2 e^{-ik_2a} = t_2 e^{ik_3a} \\ t_1 k_2 e^{ik_2a} - r_2 k_2 e^{-ik_2a} = t_2 k_3 e^{ik_3a} \end{cases}$

$\psi_{in}(x) = e^{ik_1x}$ $\quad J_{in} = Re\left\{\psi_{in}^* \frac{\hat{P}}{m} \psi_{in}\right\} = \frac{\hbar k_1}{m}$

$J_T = Re\left\{\psi_T^* \frac{\hat{P}}{m} \psi_T\right\} = Re\left[\frac{\hbar k_3}{m}\right] |t_2|^2$

$T = \left|\frac{J_T}{J_{in}}\right| = Re\left[\frac{k_3}{k_1}\right] |t_2|^2$

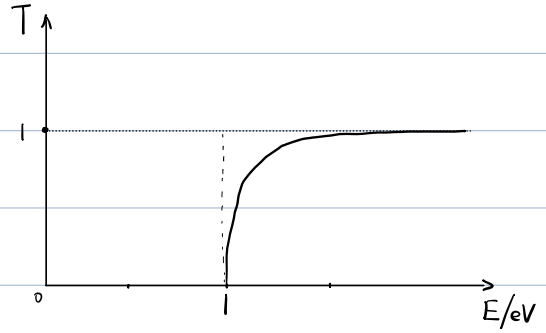
① while $E > V_2$

$T = \frac{k_3}{k_1} \left| \frac{4k_1k_2 e^{i(k_2-k_3)a}}{(k_1+k_2)(k_2+k_3) - (k_2-k_1)(k_2-k_3)e^{i2k_2a}} \right|^2 = \frac{16k_1k_2^2k_3}{\left|(k_1+k_2)(k_2+k_3) + (k_1-k_2)(k_2-k_3)e^{i2k_2a}\right|^2}$

$T = \frac{16\sqrt{E}(E-V_1)\sqrt{E-V_2}}{\left|(\sqrt{E}+\sqrt{E-V_1})(\sqrt{E-V_1}+\sqrt{E-V_2}) + (\sqrt{E}-\sqrt{E-V_1})(\sqrt{E-V_1}-\sqrt{E-V_2})e^{i2k_2a}\right|^2}$

② while $E < V_2$.

$T = Re\left[\frac{k_3}{k_1}\right] |t_2|^2 = 0$



$t_1 e^{ik_1a} + r_2 e^{-ik_1a} = t_1 \frac{k_2}{k_3} e^{ik_2a} - r_2 \frac{k_2}{k_3} e^{-ik_2a}$

$r_2 = t_1 \frac{k_2 - k_3}{k_2 + k_3} e^{i2k_2a}$

$2 = \left(1 + \frac{k_2}{k_1}\right) t_1 + \left(1 - \frac{k_2}{k_1}\right) r_2$

$\hookrightarrow t_1 = \frac{2k_1}{k_1+k_2} + \frac{k_2-k_1}{k_1+k_2} r_2$

$t_1 = \frac{2k_1}{k_1+k_2} + \frac{k_2-k_1}{k_1+k_2} \cdot \frac{k_2-k_3}{k_2+k_3} \cdot t_1 e^{i2k_2a}$

$t_1 = \frac{\frac{2k_1}{k_1+k_2}}{1 - \frac{(k_2-k_1)(k_2-k_3)}{(k_1+k_2)(k_2+k_3)} e^{i2k_2a}}$

$t_1 = \frac{2k_1(k_2+k_3)}{(k_1+k_2)(k_2+k_3) - (k_2-k_1)(k_2-k_3)e^{i2k_2a}}$

$r_2 = \frac{2k_1(k_2-k_3)e^{i2k_2a}}{(k_1+k_2)(k_2+k_3) - (k_2-k_1)(k_2-k_3)e^{i2k_2a}}$

$\Rightarrow t_2 = \frac{4k_1k_2 e^{i(k_2-k_3)a}}{(k_1+k_2)(k_2+k_3) - (k_2-k_1)(k_2-k_3)e^{i2k_2a}}$

3. (a) wave function: $\psi(x,y)$.

In x directions: $-\frac{\hbar^2}{2m}\cdot\frac{\partial^2\psi}{\partial x^2} = E\cdot\psi$

$$\psi_x(x) = \begin{cases} A\cos kx + B\cdot\sin kx, & 0 < x < 4nm \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{cases} \psi_x(0^-) = \psi_x(0^+) = 0 \\ \psi_x(4\times10^{-9-}) = \psi_x(4\times10^{-9+}) = 0 \end{cases}$$

$\Rightarrow A = 0. \quad 4\times10^{-9}\times k_x = n_x\pi, \ n = 0, 1, 2\ldots$

$\Rightarrow k_x = \frac{\sqrt{2mE_x}}{\hbar} = \frac{n_x\pi}{4\times10^{-9}}$

$\Rightarrow E_x = \frac{n_x^2\pi^2\hbar^2}{32m}\times10^{18}$

Similarly, $E_y = \frac{n_y^2\pi^2\hbar^2}{18m}\times10^{18}$

$\Rightarrow E_n = E_x + E_y = \frac{\pi^2\hbar^2\times10^{18}}{m}\cdot\left(\frac{n_x^2}{32}+\frac{n_y^2}{18}\right)$, $n_x = 0, 1, 2, \ldots$, $n_y = 0, 1, 2, \ldots$

b) Similarities: The energy is Quantized, dependence on the mass of particle and well dimensions.

Differences: There are two quantum numbers, different (n_x, n_y) may Corrospond to the same E_n

4. $\delta(x) = \begin{cases} 1, & x\neq0 \\ \infty, & x=0 \end{cases}$ $\int_{-\infty}^{+\infty}\delta(x)\,dx = 1$

$$\Rightarrow \int_{-\infty}^{+\infty}\delta(x)\,dx = \int_{-\infty}^{0^-}\delta(x)\,dx + \int_{0^-}^{0^+}\delta(x)\,dx + \int_{0^+}^{+\infty}\delta(x)\,dx = 1$$

$$\Rightarrow \int_{0^-}^{0^+}\delta(x)\,dx = 1$$

$$\int_{-\infty}^{+\infty}\delta(x-x_0)f(x)\,dx = \int_{-\infty}^{x_0^-}\delta(x-x_0)f(x)\,dx + \int_{x_0^-}^{x_0^+}\delta(x-x_0)f(x)\,dx + \int_{x_0^+}^{+\infty}\delta(x-x_0)f(x)\,dx$$

$$= \int_{x_0^-}^{x_0^+}\delta(x-x_0)f(x)\,dx$$

$$\Rightarrow \int_{-\infty}^{+\infty}\delta(x-x_0)f(x)\,dx = f(x_0)$$

5. (a) Assume wave function : $\psi(x)$

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx} & x<0 \\ te^{ikx} & x>0 \end{cases} \qquad k = \frac{\sqrt{2mE}}{\hbar}$$

$$\psi(0^-) = \psi(0^+) \Rightarrow 1 + r = t$$

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\cdot\psi(x)$$

$$-\frac{\hbar^2}{2m}\int_{0^-}^{0^+}\psi''(x)\,dx + \int_{0^-}^{0^+}V(x)\psi(x)\,dx = \int_{0^-}^{0^+}E\,\psi(x)\,dx$$

$$-\frac{\hbar^2}{2m}\left(\psi'(0^+) - \psi'(0^-)\right) + V_0\psi(0) = 0$$

$$ik\frac{\hbar^2}{2m}(t - 1 + r) = (1+r)V_0$$

boundary condition : $\begin{cases} ik\frac{\hbar^2}{2m}(t-1+r) = (1+r)V_0 \\ 1+r = t \end{cases}$

(b) $J_{in} = Re\left\{e^{ikx}\cdot\frac{-i\hbar\frac{\partial}{\partial x}}{m}\cdot e^{-ikx}\right\} = Re\left\{\frac{\hbar k}{m}\right\}$

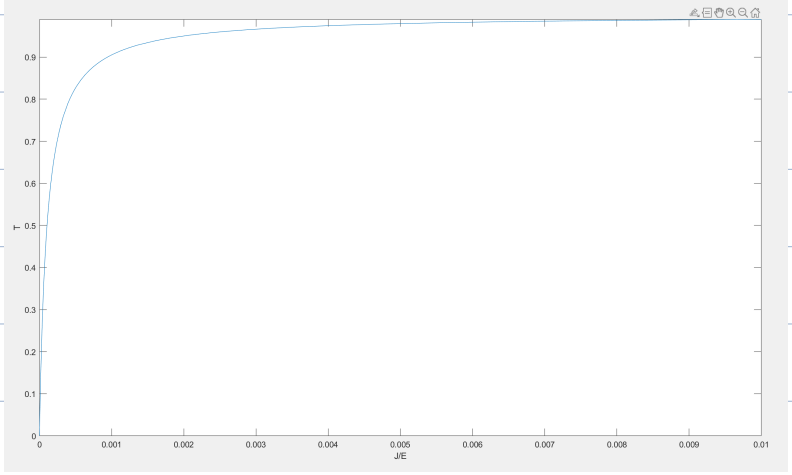
$$J_T = Re\left\{t\cdot e^{ikx}\cdot\frac{-i\hbar\frac{\partial}{\partial x}}{m}\cdot t\cdot e^{-ikx}\right\} = Re\left\{\frac{\hbar k}{m}\right\}\cdot|t|^2$$

$$T = \left|\frac{J_T}{J_{in}}\right| = |t|^2$$

$$\begin{cases} ik\frac{\hbar^2}{2m}(t-1+r) = (1+r)V_0 \\ 1+r = t \end{cases}$$

$$t = \frac{1}{1 + i\cdot\frac{V_0 m}{k\hbar^2}} = \frac{1}{1 + i\cdot\frac{V_0}{E}\sqrt{\frac{m}{2E}}}$$

$$T = |t|^2 = \frac{1}{1 + \frac{V_0^2 m}{2\hbar^2 E}}$$



6. (a) In region I:

$$-\frac{\hbar^2}{2m}\psi_{(x)}'' + V_1\psi_{(x)} = E\psi_{(x)}$$

$$\psi_{(x)} = A_1 e^{k_1 x} \quad , \quad k_1 = \frac{\sqrt{2m(V_1 - E)}}{\hbar}$$

In region II:

$$-\frac{\hbar^2}{2m}\psi_{(x)}'' = E\psi_{(x)}$$

$$\psi_{(x)} = A_2 \cos kx + B_2 \sin kx \quad k_2 = \frac{\sqrt{2mE}}{\hbar}$$

In region III. $\psi_{(x)} = 0$

$$\Rightarrow \psi_{(x)} = \begin{cases} A_1 e^{k_1 x} & x < 0 \\ A_2 \sin k_2 x + B_2 \cos k_2 x & 0 < x < a \\ 0 & x > a \end{cases}$$

b): $\begin{cases} \psi_{(a^-)} = \psi_{(a^+)} \\ \psi_{(0^-)} = \psi_{(0^+)} \\ \psi'_{(0^-)} = \psi'_{(0^+)} \end{cases}$

$$\Rightarrow \begin{cases} A_1 = B_2 \\ A_1 k_1 = A_2 k_2 \\ A_2 \sin k_2 a + B_2 \cos k_2 a = 0 \end{cases}$$

$$\Rightarrow \tan k_2 a = -\frac{k_2}{k_1}$$

c) $\tan k_2 a = -\frac{k_2}{k_1}$

$$\Rightarrow \tan\left(\frac{\sqrt{2mE}}{\hbar} \cdot a\right) = -\sqrt{\frac{E}{V_1 - E}}$$

The equation has two solutions: $E_1 = 0.0135$ eV, $E_2 = 0.0338$ eV